

Equivalence Class Partitioning (ECP)

Equivalence Class Partitioning is a manual testing technique that divides the task into valid and invalid class. In this technique, first the set of condition is divided into partition. It can be applied to many types of testing like unit testing, integration testing. It tests one value per class and covers maximum functionality of the code.

Eg 1: The age range of a person is 1 - 75. So, the valid portion is 1 to 75 years, whereas, the invalid portion is less than 1 year or more than 75 years.

Eg 2: In a mobile banking app, lets say Esewa as a scenario, it only accepts upto Rs. 1,00,000 at a single time. So, the test case is Re. 1 to Rs. 1,00,000 is valid, whereas, less than Re. 1 (like 0.75) or more than Rs 1,00,000 (like 1,20,000) is invalid.

Boundary Value Analysis (BVA)

Boundary Value Analysis (BVA) is a manual testing technique in which the test is performed based on the edges of input ranges. It focuses on the boundary values like maximum, minimum, more than maximum or less than maximum values. It tests the limit of ranges where errors are most likely to be occurred.

Eg 1: The age range is 1 - 100 years. So, BVA focuses on the values like 1, 100, less than 1 (minimum value) i.e. 0 or 0.5 and more than 100 (maximum value) i.e. 101.

Eg 2: In a mobile banking app, the transaction amount is Rs 2 to Rs 500. So, it tests the boundary like 1, 2, 3 and 499, 500, 501.

Decision Table Testing.

Decision Table is a technique used to test the system based on input and its corresponding output. This technique tests by making a table that consists of input and expected output. It has two aspects; conditions and actions. These two are matched to figure out the result. It has complete code coverage of input test cases and doesn't overlook any test cases.

Eg 1: A departmental store like Big Mart or Sales Berry have 2 different segments of actions and conditions.

Conditions (The total price of products is above Rs 1000 (Y/N), The buyer has membership (Y/N))

Actions (Apply 20% discount, No discount)

In this scenario, if test 1 has both two conditions yes, then the 20% discount is applied. If test 2 has only 1 or neither of the conditions yes, then the discount is not applied.

Eg 2: In a college like Islington College or Herald College:

Conditions (Student got good grades (Y/N), Student got good attendance (Y/N))

Actions (Apply 50% scholarship, No scholarship)

In this scenario, if test 1 has both the conditions yes, then scholarship is given. If test 2 has only one of the conditions yes, then the scholarship is not given.

State Transition Testing

State Transition Testing is a technique in which changes made in input conditions cause state changes or results in transition of the state as the name suggests. Here, the input condition and output condition is very much interrelated. This technique is used to check the change of

the in the application. It is based on the events and inputs and is useful when the system is based on the events.

Eg 1: In a security system like HiGain HRS Login, there are 3 states (login, logout, account locked) and 2 inputs (correct password, incorrect password). If the user enters the correct password, the state transits from logout to logged in. But if the user enters incorrect password (lets say for 3 times in a row), the state transits from logged out to account locked.

Eg 2: In a real life automatic-car scenario, idle and movement are the events, whereas accelerator and brake are inputs. So, if the driver press the number accelerator, the car moves and transits from idle to movement and if the driver press the brake, the car stops and the state transits from movement to idle.

Use of testing techniques in real life

Testing is a very important term not only inside the QA bracket but all over things in every sector across different region. Testing is a process that finds errors before the confirmation, validation or before setting the green to the product not just in computing but every sectors. Testing ensures durability, quality, adaptability of the product or service. In the QA filed, use of testing technique is equally important as developing and coding. Testing is done throughout many stages of a project development lifecycle.

Testing techniques like BVA, ECP, State Transition, Cause-Effect Graphing, Decision Table have different roles and are equally important in the respective task. These techniques helps to thoroughly analyze the problem and breaks down into simpler form so that it can be solved and errors can be found.

- ECP tests valid and invalid portions that can be applied in life like age range, amount range or any digits with the starting and ending point.
- BVA tests boundaries that can be applied in ATM, banking, schools or any other sectors.
- State Transition tests the security system which is very important in today's world as many malicious acts are growing.
- Cause-Effect Graphing tests associates the input and output in a graphical form. Can be applied in real life to calculate the expected result of certain events.
- Decision Table is similar as Cause effect. It tests in a structured table format.

Hence, use of testing techniques can be seen in apart from technical field too.